CallMinder™ — the development of BT's new telephone answering service

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The CallMinder™ service, or 'answering machine in the network', is proving to be very popular with BT's customers. The introduction of the service broke new ground in being the first large-scale deployment of interactive voice platforms within BT's network. This paper describes the service, the platform architecture and the development and deployment activities.

1. Introduction

CallMinder is one of BT's new Select Services. It combines the features of a normal telephone answering machine with the ability to answer calls on busy. The service is aimed at personal customers and single line business customers.

Approximately 40% of all calls over BT's network either go unanswered or receive busy tone. By taking a message, CallMinder provides a valuable customer service and also improves the call completion rate and hence network efficiency.

There are significant advantages for the user of the Callminder service:

- no customer premises equipment required,
- simple access from home or remote telephone to retrieve messages or configure the service,
- simple and friendly dialogue with voice or TouchTone [1] commands,
- no premium charged for usage — standard network tariffs for message recording, free access from home for message retrieval,
- the service takes messages both on no-reply and also on busy — this is a key differentiator from answering machines, particularly with the rapid introduction of data services such as fax and Internet, where a telephone line can be tied up for long periods, and yet CallMinder will continue to take messages for the user,
- reliable and capacious message storage.

Initial human factors trials of CallMinder applications showed that usability was key to achieving market penetration for the service. For most users, CallMinder would represent their first experience of communicating with a machine.

Following a series of small-scale human factors trials, a marketing trial with 1000 customers commenced in Orpington in 1993. This trial was based on prototype hardware and produced valuable data on usability, usage patterns and service dimensioning.

The data was used to develop a business case for the national roll-out of a CallMinder service and refine the user interface still further. The subsequent roll-out programme included a pilot, and a limited launch within the London area prior to a national roll-out in 1995.

2. Main features

The CallMinder application offers both a standard and an enhanced service. The standard service is targeted at residential customers and is described below. The enhanced service is targeted more at small businesses, providing greater mailbox size and greeting message length, and the simultaneous answering of a greater number of calls.

[1] TouchTone™ known technically as dual tone multiple frequency (DTMF), are the tones sent when the keys are pressed on a tone dialling telephone.

2.1 Call answering

In addition to answering calls on 'ring tone no reply', Callminder answers up to three calls simultaneously on busy. This feature, not available from traditional answering machines, can be particularly effective for small businesses in managing their call flow and capturing enquiries.

Once the call is answered the caller can be greeted with either a personal message from the CallMinder owner, or a pre-recorded BT greeting message.

When the caller leaves a message, they have the opportunity to listen to it and re-record it if they wish — very useful if they have said something they had not quite meant to say!

They can repeat this re-recording process up to four times leaving a message of up to 5 min in length.

2.2 Message retrieval

CallMinder owners can retrieve their messages in one of two ways:

- message retrieval from their own telephone by dialling a short code access number (1571),
- message retrieval from a remote telephone by dialling their own telephone number and entering a 4-digit PIN [1] number when the call is answered by CallMinder — this interrupts the CallMinder greeting message and switches the application into message retrieval mode.

In addition to being able to play, repeat and save, or remove their messages, CallMinder owners can:

- review and change their personal greeting,
- change the 'ring tone no reply' period before CallMinder answers their calls to one of four settings : zero, short, medium or long, corresponding to 0, 12, 21 or 30 sec,
- change their PIN number (available from local message retrieval only as an additional security measure),
- change to a 'fast track' message retrieval dialogue, targeted at experienced users,
- change the addressee setting, which changes the caller dialogue to ask the name of the person who is calling.

[1] A 4-digit numeric password configurable by the owner.

2.3 Message waiting indication

CallMinder owners are informed that a message is waiting for them by two methods:

- changing an owner's dial tone to a distinctive 'stuttered' tone,
- ringing the owner after messages have been taken on 'busy' and informing them that a message has been taken. The service will make up to four attempts to ring the owner, at 2, 5, 15 and 30 minutes after the message was taken.

2.4 Speech processing

A novel feature of CallMinder is that owners can retrieve their messages and change the features of the service by instructing the system using speech recognition. This provides a more natural interface than traditional TouchTone dialogues, and can be used on any telephone including the older 'pulse dial' types. To facilitate customer choice, TouchTone detection is provided in parallel to the speech recognition, and a caller may even use a mixture of the two in the same call.

The speech recognition used is speaker independent, allowing CallMinder to cater for a wide range of UK dialects without any user-specific training. The following words can be used:

'yes', 'no', 'yes please', 'no thank you';
'zero', 'short', 'medium', 'long';
'zero', 'oh', 'nought', 'one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine'.

Recorded messages are stored at 8 kbit/s, using a state-of-the-art encoding algorithm developed at BT Laboratories. This algorithm provides high-quality recordings using a bit rate which is significantly lower than the 32 kbit/s storage rate more commonly used. In subjective assessments against a range of analogue and digital answering machines, the 8 kbit/s encoding was found to provide higher quality recordings. The lower bit rate reduces the hard disk storage required for CallMinder and so lowers the platform cost.

3. The architecture

3.1 Overview

The CallMinder platform architecture is shown in Fig 1. The platform used is the interactive speech applications platform (ISAP), described in Twist et al [1].

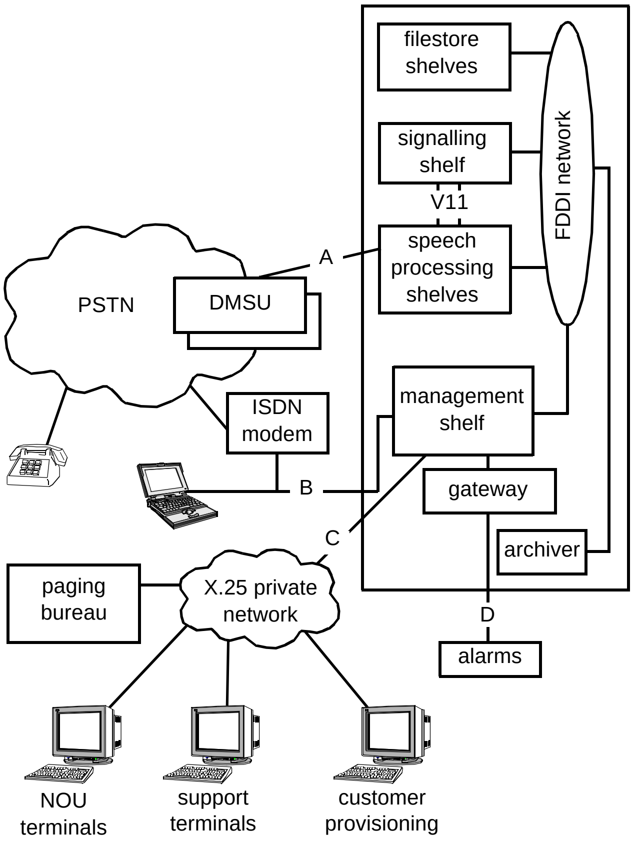

Fig 1 CallMinder platform architecture.

The platform is fully integrated into the network, connecting directly to the Digital Main Switching Units (DMSUs)[1] using BT standard C7 network signalling.

The platforms are controlled remotely from existing Network Operations Units (NOUs). Alarms are also extended remotely, via the BT Network Operations Management System (NOMS1) alarm system. Customers are loaded automatically — from taking the order on BT's Customer Services System (CSS), through to the Operations and Maintenance Centres System (OMC) and thence to the CallMinder platform. Service data is downloaded overnight to existing management information systems.

3.2 Interfaces

There are four primary interfaces to the platform, shown as A (network interface), B (data collection interface), C (operations interface) and D (alarm management interface) in Fig 1.

[1] The backbone switching units in BT's UK network.

Network interface

Each ISAP is connected to two or three digital main switching units. Each of the DMSUs connects via a single CCITT No 7 (C7) route, consisting of two links into the ISAP. Traffic can be routed to the platform via any of the attached DMSUs, and via either link of each DMSU to ISAP route. This therefore provides resilience against network routeing, DMSU or platform failure, as traffic may route around such a problem.

Message deposit calls are routed to the platform by means of a divert which is set on the owner's line at their local exchange. The C7 information routed to the ISAP contains the caller's number, the owner's number (last diverted line identifier) and a service code to indicate to the ISAP whether the call diverted was on 'busy' or 'ring tone no reply'.

The C7 nodal end-to-end data protocol is used to change the owner's 'ring tone no reply' period and to apply and revoke changed dial tone to their line.

Data collection interface

Statistics collection is performed over an ISDN2 interface. Collection is undertaken overnight using standard file-transfer utilities.

Statistical data collected over the course of a day is downloaded to the Tinsley Park Computer Centre where it is loaded into an Oracle database. A management information system allows subsequent access to this data over BT's multiprotocol router network.

Operations interface

Each ISAP has an X.25 interface to BT's private data network. This network allows closed user group access to the platform for system and service management purposes.

- Automated customer provision

  The (automated) customer provision process overcomes the problems of errors, cost and throughput associated with manual systems. When the customer's request for CallMinder is entered into the Customer Services System, it is forwarded electronically to the appropriate OMC. The OMC software first creates the mailbox on the ISAP. If this is succesful then the OMC invokes the administration-controlled diversion on the customer's local exchange. The OMC interface to the ISAP is via X.25 over BT's private network. The interface can support up to 1000 customer transactions per day on each ISAP and can support:

— customer provision,

— customer audits,

— customer cessation,

— change to customer's directory number,

— alteration of customer's CallMinder product.

- System management

  System management is menu driven using a character based system. There are five levels of support:

  — system management, allowing:

  access to error logs,
  ability to reboot/shutdown system components (e.g. speech shelves),
  C7 maintenance actions (e.g. restart link),
  alarm management,

  — security, allowing:

  account management,
  X.25 management,
  password management,

  — support, allowing:

  platform operating system access,
  access to system and service management and data collection commands,

  — data collection, allowing:

  access to platform statistics (e.g. traffic levels),

  — service management, allowing:

  access to customer provision/cessation facilities,
  service statistics (e.g. number of customers, usage).

Alarm management interface

Alarms are forwarded to the NOMS1 alarm-reporting system via a gateway system or via an alarm-collection unit. These standard BT alarm boxes collect alarms in the form of signals on input wires (0 = active, 1 = inactive) and send the outputs via a modem to the NOMS1 system. Alarms are prioritised into one of four levels:

FATAL, ERROR, WARNING or INFORMATION

Alarms can be viewed and reset via the system management menus.

4. Dimensioning

The CallMinder system has been designed to be easily extended. The initial installation is dimensioned to meet the first year traffic forecasts. Individual platforms can be extended up to two to three times their initial capacity, and additional platforms can be installed in current or new sites.

4.1 Network dimensioning

Based on the forecast traffic, and assuming an even distribution across the UK population, a network costing analysis showed that the optimum configuration would be for each CallMinder platform to serve three network DMSUs, except in London where the optimum configuration would be four platforms serving the eight DMSUs. Thus the total number of platforms required was 21, spread nationally and located at 19 sites, as shown in Fig 2.

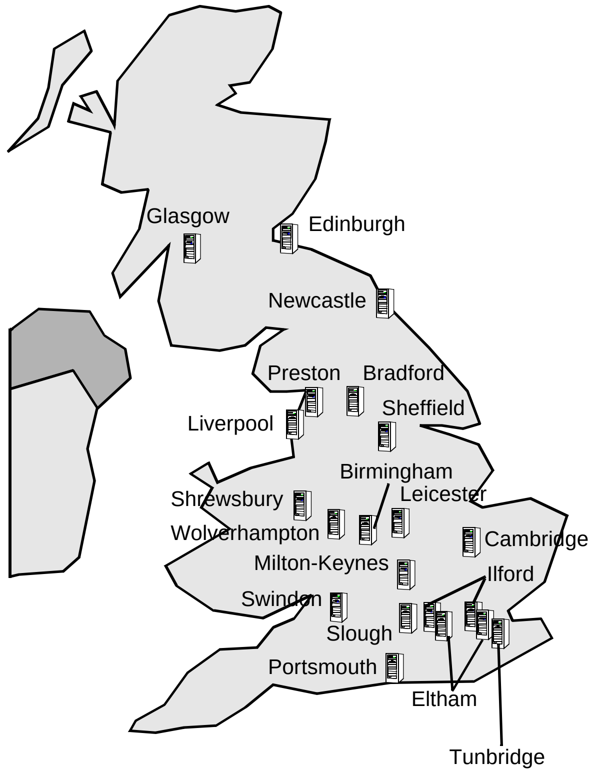

Fig 2 CallMinder UK sites.

4.2 Platform dimensioning

The forecast take-up of the service required the total initial installed capacity to support approximately 900 000 users. This capacity was spread evenly between the sites — the four London ISAPs were dimensioned to support 50 000 customers, with the remaining ISAPs dimensioned at 40 000 customers (giving a total capacity of 880 000 users).

Platform dimensioning took account of many factors, including:

- number of mailbox owners,
- message storage capacity,
- call arrival rate,
- typical call duration,
- distribution of call types — message storage, message retrieval, user configuration,
- typical use of speech resources (e.g. usage of speech recogniser, speech encoder and decoder),
- outgoing calling rate (e.g. for message waiting indication calls and setting stutter dial tone),
- service provisioning rate (e.g. number of new users per hour, day),
- typical data collection usage,
- typical system and service management usage.

This information was derived from the business case forecasts and from data from the marketing trials.

As reliability and availability were identified as key market requirements, the CallMinder platforms have been designed to be resilient to individual component failure. Some of the resilient features included are:

- disk mirroring for all data storage,
- N+1 redundancy for speech processing components,
- hot standby of network signalling components,
- support for dynamic signalling link re-routeing,
- uninterruptable power supplies, and N+1 redundancy of power supply components,
- dual attached optical fibre ring as platform local area network,
- multiple and diverse network and operational connections.

The ISAP architecture is described in more detail in Twist et al [1]. For information, the configurations used for the initial installations are as shown in Table 1.

Table 1 Initial installation configurations.

Shelf type	Number of Shelves	
	London site	Other sites
Speech processing	6	5
Filestore	4	3
Management	1	1
C7	1	1
Archiver	1	1

5. Managing the development

5.1 Introduction

Following the evaluation of tenders for voice platforms from leading international suppliers, the ISAP bid by Ericsson Ltd was chosen on the grounds of cost, functionality and future enhancements. The ISAP had originally been developed by BT Laboratories and subsequently licensed to Ericsson Ltd, who would also act as the prime contractor for delivery of the CallMinder service. BT Laboratories would act as a subcontractor for the delivery of the application on a fixed-price basis.

The CallMinder roll-out development was planned in three phases:

- a single platform pilot service — to prove the platform, operational feasibility and marketing case,
- a limited roll-out to the four London sites to gain installation and operational experience (the service would not be officially launched at this stage),
- a full roll-out to all the national sites, and a national launch.

5.2 Pilot service

The objective of the CallMinder pilot service was to prove the new ISAP platform technology, and trial the network integration and operational processes proposed. The service was based on a single platform sited at Ealing, with 1000 customers transferred from the 1993 Orpington market trial.

The pilot development followed a similar life cycle to that described below, although simplified to reflect the single-site trial nature of the delivery.

A key element of the pilot development was the transfer of ISAP manufacture and hardware design and support knowledge to Ericsson Ltd.

In order to reduce launch timescales, the pilot development overlapped the roll-out development to some extent, requiring careful management of the two projects.

The pilot development commenced in June 1993, with the service launched on schedule in March 1994. The platform remained in service until deployment of the London roll-out service in October 1994. As well as providing an operational test bed, the pilot service was also used to fine-tune the marketing campaign for national launch, such as investigating customer price sensitivity.

5.3 Roll-out delivery structure

The roll-out of the CallMinder service was dependent on the delivery of a number of new components:

- ISAP Build 4.0 — the core platform hardware and software build,
- Visage Build 3.2 — the service creation tool for dialogue development,
- CallMinder Application,
- OMC Build Q — customer provisioning software,
- CSS Release 23 — customer order-handling software,
- AXE10 NEP — Ericsson local exchange software,
- System X 4900 Build — GPT local and main exchange software.

In addition, new customer service and operational practices needed to be developed across BT.

The integration of such a large number of new components and processes carried considerable risk, which required careful management. Both CSS and OMC changes were delivered through existing programmes, although their activities were co-ordinated with the CallMinder application development through an umbrella programme.

The networks and systems area within BT acted as the delivery agents and managed contracts let to GPT [1] (for the System X switch changes) and Ericsson Ltd. As well as acting as prime contractor for the delivery of the CallMinder platform and application, Ericsson were also responsible for AXE10 switch changes. Ericsson subcontracted the platform software and CallMinder application development to BT Laboratories. The platform hardware design and manufacture were transferred to Ericsson as part of the pilot delivery, although BT Laboratories provided design support as necessary.

The application development contract was a fixed-price, fixed-timescale contract, involving various parts of BT and Ericsson in a relationship that, in practice, worked very well.

[1] GEC Plessey Telecommunications — suppliers of the System X main and local network switches to BT.

5.4 The application development

The CallMinder application software runs on the ISAP, and manages the customer dialogue, the storage and retrieval of messages, customer provisioning, statistics and element management. There are several components to the application, running on different processors in the platform and managing the platform interfaces as described earlier in this paper. The application development project was combined for the London and national roll-out deliveries, and split into the following seven stages.

Stage 0: Project initiation and support

The project initiation stage included the production of project proposals, the financial case and initial project plan, and the allocation of suitable resource to the project.

Stage 1: Definition

Key deliverables from the definition stage were:

- statement of requirements,
- compliance response,
- requirements specification,
- project plan.

The requirements capture process derived a single statement of requirements, compliance response and requirements specification from eight separate and often conflicting source documents, as shown in Fig 3.

Stage 2: Design

The design stage followed standard processes. The design of the dialogue was more involved, as it included the generation of a dialogue prototype for human factors trials, and incorporation of feedback from those trials back into the application. The dialogue was designed in accordance with the BT Voice Applications Style Guide [2]. Design deliverables included:

- architectural design,
- component specifications and designs,
- dialogue design.

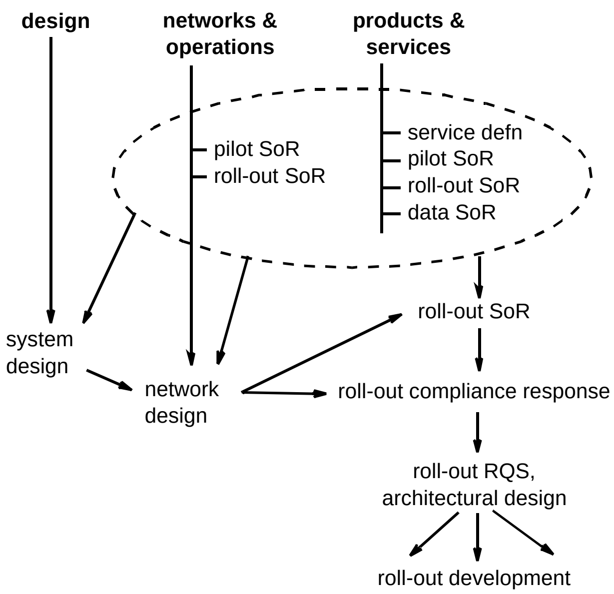

Fig 3 Requirements map.

Stage 3: Implementation

The implementation stage included module code and test, and integration and acceptance of application and ISAP software into a stable system prior to commencement of the test stage.

Stage 4: Test

The test stage was the culmination of VV&T activities spread over the entire life cycle of the project, which included:

- requirements testability analysis,
- requirements tracking,
- design reviews,
- documentation reviews,
- code walk-throughs,
- test harnesses,
- call generator (specifically commissioned for this project),
- signalling test bed,
- commissioning of a full-sized reference model.

The test stage was very complex and had to meet requirements including system testing, documentation validation, network interconnect validation testing (IVT), network interworking testing (IWT), system integration testing and formal development acceptance testing. The IWT activity included the integration of the ISAP with OMC, CSS, AXE10 and SystemX elements in the network integration facility (NIF). Performance and resilience testing were included in both system test and IWT. Key milestones from the test stage were:

- BT network interconnection certification,
- BT formal development acceptance.

Without interconnection certification, the system would have been prohibited from connection into BT's network.

Stage 5: London roll-out release

The approved platform and application software was released to Ericsson for distribution to CallMinder sites. In parallel with the software development, Ericsson had been manufacturing, installing and commissioning ISAPs in each of the sites around the country.

One of the main risks to successful roll-out was identified as training. To reduce the risk, a joint BT and Ericsson team, in addition to the planned training events, acted both as installation and commissioning agents, and as first-line support to the novice operations teams. This proved critical to the success of the project.

Stage 6: National roll-out test and release

The national roll-out software was essentially the same as that for the London release, but needed to be validated against different AXE10, SystemX and OMC builds.

The national release followed the same process as for the London roll-out. The special joint team again proved invaluable as they toured the country installing and commissioning the systems and supporting the local maintenance teams.

6. Roll-out

The CallMinder roll-out development was delivered on schedule, and in fact installation was achieved slightly earlier than planned. After a series of trial transfers to ensure no disruption to service, customers were transferred from the Ealing pilot system to the four London roll-out sites in October 1994. These sites were then upgraded to national roll-out software with all national roll-out sites operational by mid-December 1994.

Before the service could be launched, reliable operation and operational and customer services needed to be proven on all sites. This process highlighted the difficulties of the fast deployment of such a complex and advanced system. Some of these roll-out issues were:

- the scale of the roll-out, with a total of 21 platforms to be simultaneously installed and launched from 19 sites nationwide — severely stretching management, reporting and support processes,
- minor operational and network data configuration differences across the country — causing service problems which had not been encountered during the testing phase or London roll-out,

- a large-scale training programme which had to be fitted around existing commitments,
- the roll-out and subsequent launch, often requiring the resolution of conflicting priorities across the business.

One of the critical areas for service launch was the customer order taking and provisioning process. Because of the large number of people and systems, much time had to be spent bringing this process on line.

7. Launch

Following national roll-out, and prior to the service launch which was delayed due to operational difficulties and the discovery of a faulty batch of hard disks (used for message storage on the ISAP platforms), some experience of platform operation was provided by 1000 BT people being recruited to take up the service. The service was formally launched nationwide in May 1995. Currently, take-up is approximately 500 000 customers nationwide, and BT is already extending platforms in certain locations.

Feedback has been extremely positive, with customers appreciative of the service's ability to answer calls on 'busy' and its user friendliness.

8. The future

The immediate future for CallMinder is likely to see the current platforms extended to cope with demand.

In the medium term, enhancements are planned to:

- allow better feature interworking with other select services, particularly call diversion and call waiting,
- allow message retrieval on message waiting indication calls,
- allow different personal greetings for 'ring tone no reply' and 'busy'.

Other longer term enhancements are likely to see different product sets of CallMinder aimed at particular market sectors such as students and small businesses. Longer term features being considered include:

- fax store and forward,
- memo facility,
- message waiting indication to use ring back when free,
- call screening,
- paging message-waiting indication.

9. Conclusion

This development of the CallMinder service has provided many technical, commercial, operational, management and marketing challenges. The successful delivery of CallMinder into service is a tribute to the many teams across BT and its suppliers that have worked together to overcome these challenges, which included:

- use of the interactive speech applications platform,
- development of system interfaces,
- network and platform dimensioning to meet customer demand,
- joint development between BT and Ericsson.

With service take-up rates continuing to rise, and good customer feedback, the future of the CallMinder service is looking secure.

References

1 Twist P, Briggs T, Morgan K and Hughes P: 'ISAP — delivering the goods', BT Technol J, 14, No 2, pp 35—42 (April 1996).

2 'Voice Application Style Guide', BT internal publication, a summary of this guide has been published as BT publication: 'Now You're Talking... Voice Services', Ref No 367 (1994).

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He was the project manager for the CallMinder application development, and currently manages the multimedia platforms unit at BT Laboratories.

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